Differential Equations

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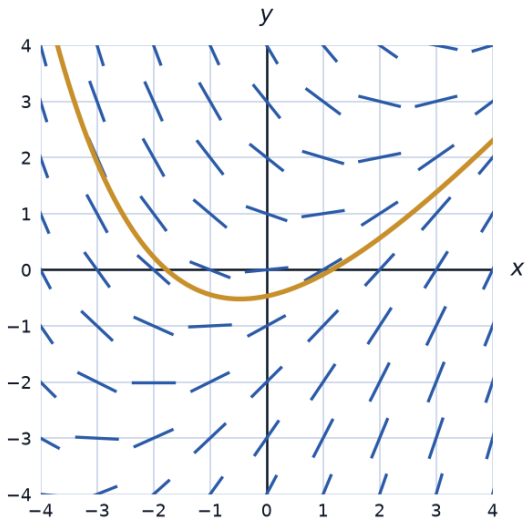
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Solving by separation of variables

Solve the differential equation $\frac{dy}{dx} = \frac{x}{y}$, given that $y = 3$ when $x = 0$.

2 [Foundational] • Target: Solving by separation of variables

Solve $\frac{dy}{dx} = 2xy$, given that $y = 5$ when $x = 0$.



3 [Foundational] • Target: Modeling with differential equations

A population P grows at a rate proportional to its current size, modeled by $\frac{dP}{dt} = kP$. If the population doubles every 5 years, find k, to three decimal places.

4 [Foundational] • Target: Modeling with differential equations

Newton's law of cooling states $\frac{dT}{dt} = -k(T - T_a)$, where T_a is ambient temperature. Solve this differential equation for $T(t)$ in terms of the initial temperature T_0.

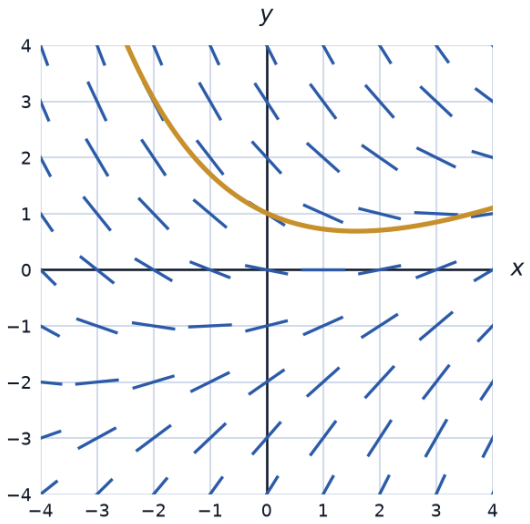
5 [Foundational] • Target: First-order linear differential equations

Solve $\frac{dy}{dx} + 2y = 6$ using an integrating factor, given $y = 1$ when $x = 0$.

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6 [Foundational] • Target: Euler's method for numerical solutions

Use Euler's method with step size $h = 0.5$ to estimate $y(1)$ for $\frac{dy}{dx} = x + y$, given $y(0) = 1$, showing each step.



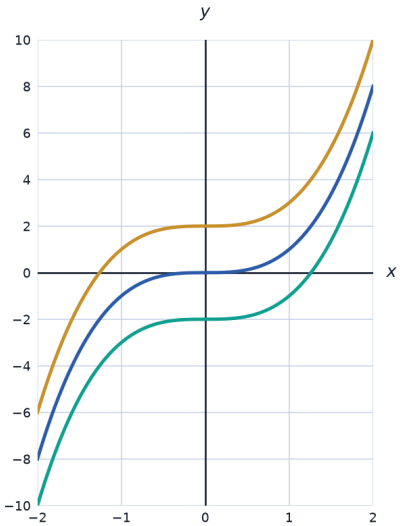
7 [Foundational] • Target: Second-order context: simple harmonic motion

The differential equation $\frac{d^2x}{dt^2} = -\omega^2 x$ describes simple harmonic motion. Verify that $x(t) = A\cos(\omega t)$ is a solution.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Family of solution curves and particular solutions

Find the general solution of $\frac{dy}{dx} = 3x^2$, and explain what the arbitrary constant represents geometrically.



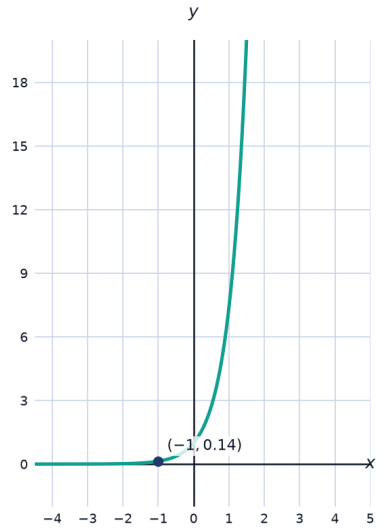
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9 **[Intermediate]** • **Target: Family of solution curves and particular solutions**

A tank drains such that $\frac{dV}{dt} = -k\sqrt{V}$ for some constant $k > 0$, where V is the volume of water. Solve this differential equation for $V(t)$, given $V(0) = V_0$.

10 **[Intermediate]** • **Target: Verifying a solution to a second-order differential equation**

Verify that $y = e^{2x}$ is a solution of the differential equation $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = 0$.



11 **[Intermediate]** • **Target: Slope fields and qualitative solutions**

A differential equation is given by $\frac{dy}{dx} = x - y$. Find the gradient of the solution curve at the points $(0, 0)$, $(1, 0)$, and $(1, 1)$, and explain what a slope field represents.

12 **[Intermediate]** • **Target: A logistic growth model**

A population P grows according to the logistic differential equation $\frac{dP}{dt} = kP(10 - P)$. Using the partial fraction decomposition $\frac{1}{P(10-P)} = \frac{1}{10P} + \frac{1}{10(10-P)}$, solve for P in terms of t, given $P(0) = 1$ and leaving k as an unknown constant.

13 **[Intermediate]** • **Target: Checking the order and degree of a differential equation**

State the order of each of the following differential equations: (a) $\frac{dy}{dx} = 3x^2 - 2y$, (b) $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} - y = 0$, (c) $\frac{d^3y}{dx^3} = \sin x$.

14 **[Intermediate]** • **Target: Using Euler's method with a smaller step size**

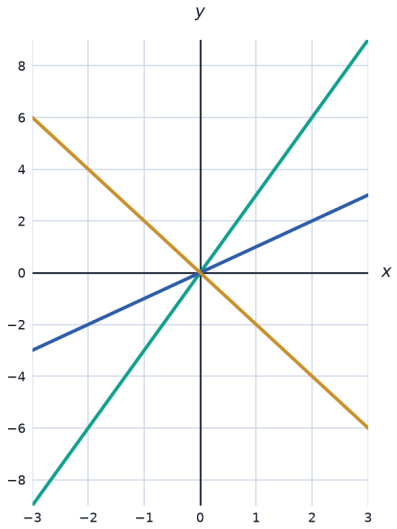
Use Euler's method with step size $h = 0.25$ to estimate $y(0.5)$ for $\frac{dy}{dx} = 2x$, given $y(0) = 1$, showing each step, and compare with the exact solution $y = x^2 + 1$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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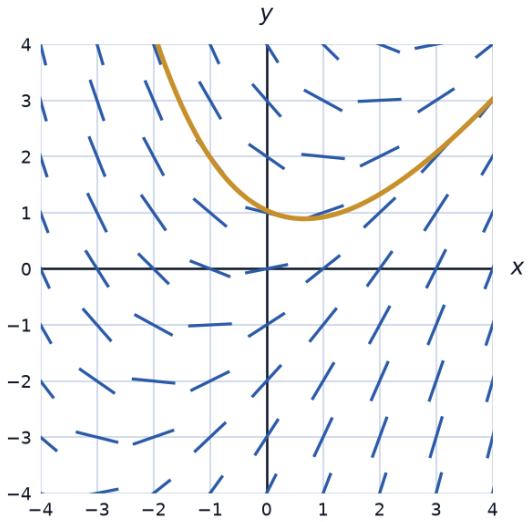
15 **[Challenging]** • **Target: Using Euler's method with a smaller step size** • **Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals**

Solve the differential equation $\frac{dy}{dx} = \frac{y}{x}$ for $x > 0$, given $y = 6$ when $x = 2$, and describe the family of curves this represents.



16 **[Challenging]** • **Target: Using Euler's method with a smaller step size** • **Cross-Lesson: 4.02 The Poisson Distribution**

A cup of coffee at 90° C is placed in a room at ambient temperature 20° C. Using Newton's law of cooling, $T(t) = T_a + (T_0 - T_a)e^{-kt}$, and given that after 10 minutes the coffee has cooled to 60° C, find the value of k to three decimal places.



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17 [Challenging] • Target: Using Euler's method with a smaller step size • Cross-Lesson: 5.01 Further Integration Techniques

This question has three parts. A tank initially contains 200 litres of pure water. Salt water containing 3 grams of salt per litre flows in at a rate of 2 litres per minute, and the well-mixed solution flows out at the same rate, so the volume stays constant at 200 litres. Let $S(t)$ be the amount of salt (in grams) in the tank at time t minutes. (a) Explain why $\frac{dS}{dt} = 6 - \frac{S}{100}$. (b) Solve this differential equation for $S(t)$, given $S(0) = 0$. (c) Find the amount of salt in the tank after 50 minutes, to the nearest gram.

18 [Challenging] • Target: Family of solution curves and particular solutions • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series

Level 1 · Foundations Find the particular solution to $\frac{dy}{dx} = \frac{3x^2}{y}$ subject to $y(0) = 4$.

19 [Challenging] • Target: Verifying a solution to a second-order differential equation • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

Solve the logarithmic equation $\log_3(x+6) + \log_3(x) = 3$ and eliminate any extraneous solutions.

20 [Challenging] • Target: Slope fields and qualitative solutions • Cross-Lesson: 6.01 Matrices (AI HL)

A curve has equation $y = 2x^3 - 9x^2 + 12x + 1$. (a) Find the coordinates of both stationary points. (b) Determine the nature of each stationary point using the second derivative $\frac{d^2y}{dx^2}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Solving by separation of variables

The function f is defined for all $x \in \mathbb{R}$ by $f(x) = x^3 - 2x^2 - kx + 12$, where k is a real constant. The Newton-Raphson method is used to find an approximation to a root of the equation $f(x) = 0$. Given that an initial approximation $x_0 = 2$ produces a first approximation $x_1 = 4$, find the value of the constant k and the exact value of the second approximation x_2.

a. $k = 5, \quad x_2 = \frac{28}{9}$

b. $k = 5, \quad x_2 = \frac{44}{9}$

c. $k = 2, \quad x_2 = \frac{14}{5}$

d. $k = 7, \quad x_2 = \frac{84}{25}$

22 [Foundational] • Target: Modeling with differential equations

First-order linear ODE solution structure: general solution = homogeneous solution +

a. particular solution

b. another homogeneous solution

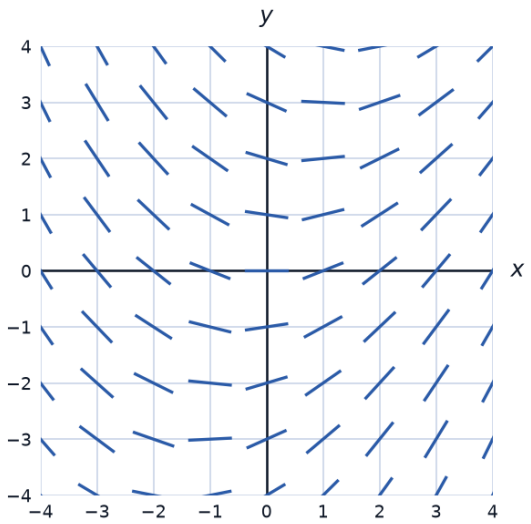
c. the initial condition

d. zero

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23 [Foundational] • Target: First-order linear differential equations

Find the particular solution of the differential equation $\frac{dy}{dx} = 2xe^{-y}$ given that $y = 0$ when $x = 0$.



a. $y = \ln(x^2 + 1)$

b. $y = \ln(x^2)$

c. $y = \ln(2x^2 + 1)$

d. $y = x^2$

24 [Foundational] • Target: Euler's method for numerical solutions

Consider the differential equation $\frac{dy}{dx} = 2x$. Which of the following is a particular solution passing through the point $(1, 3)$?

a. $y = x^2 + 2$

b. $y = x^2 + 1$

c. $y = 2x + 1$

d. $y = 2x^2 - 1$

25 [Foundational] • Target: Second-order context: simple harmonic motion

Solve $y' - \frac{y}{x} = x^2$ (linear first-order):

a. $y = \frac{1}{2}x^3 + Cx$

b. $y = \frac{1}{3}x^3 + C$

c. $y = x^3 + \frac{C}{x}$

d. $y = x^3 + Cx$

26 [Foundational] • Target: Family of solution curves and particular solutions

A second-order ODE with repeated root $r = r_1$ has general solution:

a. $y = C_1e^{r_1x} + C_2e^{-r_1x}$

b. $y = C_1e^{r_1x}$

c. $y = (C_1 + C_2)e^{2r_1x}$

d. $y = (C_1 + C_2x)e^{r_1x}$

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27 [Foundational] • Target: Verifying a solution to a second-order differential equation

A first-order linear ODE has the form:

a. $y' + P(x)y = Q(x)$

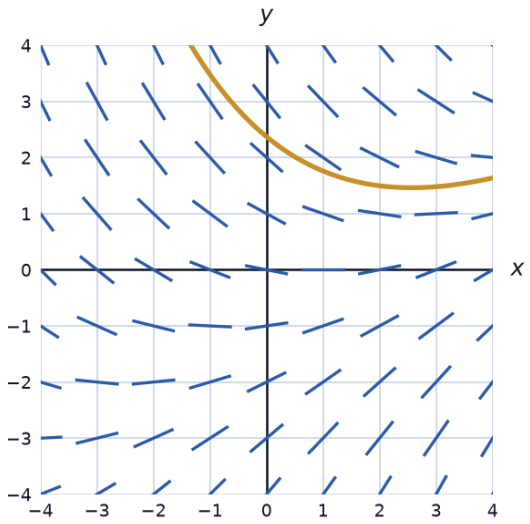
b. $y' = P(x)y^2$

c. $y'y = Q(x)$

d. $y'' = P(x)$

28 [Foundational] • Target: Slope fields and qualitative solutions

Second-order constant-coefficient ODE $y'' - y = 0$: characteristic equation:



a. $r - 1 = 0$

b. $r^2 + 1 = 0$

c. $r^2 - 1 = 0$

d. $r^2 + r - 1 = 0$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Solving a linear differential equation with an initial condition

Logistic model: $\frac{dP}{dt} = rP(1 - P/K)$. At $P = K/2$, the growth rate is:

a. zero

b. minimum

c. maximum

d. negative

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30 [Intermediate] • Target: A logistic growth model

Linear homogeneous 2nd-order ODE: if y_1, y_2 are solutions, so is:

a. y_1^2

b. $y_1 y_2$

c. y_1'

d. $y_1 + y_2$

31 [Intermediate] • Target: Checking the order and degree of a differential equation

Carrying capacity K in the logistic model: as $P \to K$ from below, growth rate $dP/dt \to$

a. 0

b. K

c. ∞

d. -1

32 [Intermediate] • Target: Using Euler's method with a smaller step size

Use Euler's method with step $\frac{1}{10}$ to approximate $y(\frac{1}{5})$ for $y' = x - y$, $y(0) = 0$.

a. 0

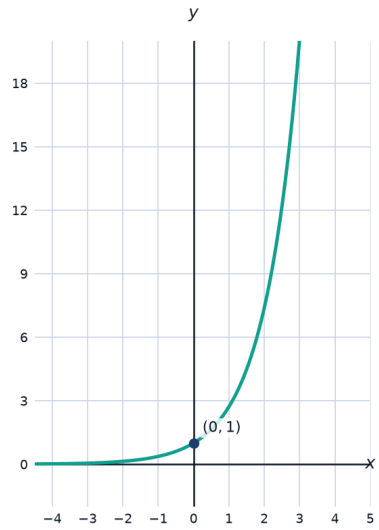
b. $\frac{1}{25}$

c. $\frac{1}{100}$

d. $\frac{11}{100}$

33 [Intermediate] • Target: Solving by separation of variables

Method of undetermined coefficients: for $y'' + y = e^x$, try particular solution:



a. $y_p = Ae^x$

b. $y_p = Ae^{2x}$

c. $y_p = Axe^x$

d. $y_p = A\sin x$

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34 [Intermediate] • Target: Modeling with differential equations

Second-order ODE with complex roots $r = \alpha \pm \beta i$: general solution includes:

a. $e^{i\beta x}$

b. $C_1 e^{\alpha x} + C_2 e^{\beta x}$

c. $(C_1 + C_2 x)e^{(\alpha + \beta)x}$

d. $e^{\alpha x}(C_1 \cos \beta x + C_2 \sin \beta x)$

35 [Intermediate] • Target: First-order linear differential equations

Reduction of order: given $y_1 = e^x$ solves $y'' - y = 0$. Second independent solution via $y_2 = v(x)e^x$:

a. $y_2 = xe^x$

b. $y_2 = e^x \ln x$

c. $y_2 = xe^{-x}$

d. $y_2 = e^{-x}$

36 [Intermediate] • Target: Euler's method for numerical solutions

Consider a particle whose velocity vector $\mathbf{v}(t) = \langle x'(t), y'(t) \rangle$ satisfies the differential equation $y'' - 2y' + y = 0$ and the initial conditions $x(0) = 1, y(0) = 2, x'(0) = 0, y'(0) = 1$. Furthermore, the particle's path is such that the magnitude of its acceleration vector is always equal to the magnitude of its velocity vector, i.e., $||\mathbf{a}(t)|| = ||\mathbf{v}(t)||$. Find the value of $x(1)$.

a. $e^{-1} + 1$

b. $2e^{-1} + 1$

c. $e^{-1} - 1$

d. $2 - e^{-1}$

37 [Intermediate] • Target: Second-order context: simple harmonic motion

Consider the exponential growth model $P(t) = P_0 e^{kt}$, where $P(t)$ is the population at time t and P_0 is the initial population. If the initial population is $P_0 = 500$ and after 10 years the population is $P(10) = 1200$, find the annual growth rate k. Round your answer to four decimal places.

a. 0.0588

b. 0.0875

c. 0.1200

d. 0.0434

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Family of solution curves and particular solutions • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series

Consider the exponential growth model described by the function $P(t) = P_0 e^{kt}$, where $P(t)$ is the population at time t, P_0 is the initial population, and k is the growth constant. If the initial population P_0 is 500 and the population after 4 years is 2000, determine the value of the growth constant k to three decimal places.

a. 0.347

b. 0.416

c. 0.301

d. 0.477

39 [Challenging] • Target: Verifying a solution to a second-order differential equation • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

Variation of parameters for $y'' + y = \sec x$: general solution involves $\ln|\sec x + \tan x|$. This appears because the particular integral of $\int \sin x \sec x \, dx =$

a. $\tan x$

b. $\ln|\cos x|$

c. $-\ln|\cos x|$

d. $-\cos x$

40 [Challenging] • Target: Slope fields and qualitative solutions • Cross-Lesson: 6.01 Matrices (AI HL)

Homogeneous 2nd-order: characteristic equation $r^2 + r - 6 = 0$. Roots:

a. $r = 2, -3$

b. $r = -2, 3$

c. $r = 1, -6$

d. $r = 3, 2$

41 [Challenging] • Target: Solving a linear differential equation with an initial condition • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)

Initial value problem: $y' = y$, $y(0) = 2$. Solution:

a. $y = e^{2x}$

b. $y = 2e^x$

c. $y = 2x + 2$

d. $y = xe^x$

42 [Challenging] • Target: A logistic growth model • Cross-Lesson: 6.03 Markov Chains (AI HL)

Second-order differential equation of a conic in polar: $r = \frac{1}{1 + e\cos\theta}$. At $\theta = 0$, $r =$

a. $1/(1 + e)$

b. $1/(1 - e)$

c. 1

d. e

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43 [Challenging] • Target: Checking the order and degree of a differential equation • Cross-Lesson: 1.01 Proof by Mathematical Induction

The equation $y' = ky$ models exponential growth when:

a. $k = 0$

b. $k < 0$

c. $k = 1$

d. $k > 0$

44 [Challenging] • Target: Using Euler's method with a smaller step size • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra

Resonance in $x'' + \omega_0^2 x = F_0 \cos(\omega_0 t)$: particular solution grows as:

a. $\sin(\omega_0 t)$

b. $e^{\omega_0 t}$

c. $t^2 \cos(\omega_0 t)$

d. $t\cos(\omega_0 t)$

45 [Challenging] • Target: Solving by separation of variables • Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity

Cauchy-Euler $x^2 y'' + xy' + y = 0$: characteristic roots $r = \pm i$. Solution:

a. $y = C_1 x^i + C_2 x^{-i}$

b. $y = C_1 \cos(\ln x) + C_2 \sin(\ln x)$

c. $y = C_1 \ln x + C_2$

d. $y = x(C_1 \cos x + C_2 \sin x)$

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